# Codebook

Circular_Window_Rebound_MAXQDA_final.mqda

26.6.2026 г.

# Code System

| | |
|---|---|
| 1 Circular strategy | 0 |
| 1.1 Maintenance | 27 |
| 1.2 Repair | 22 |
| 1.3 Component replacement | 16 |
| 1.4 Upgrade | 26 |
| 1.5 Refurbishment | 23 |
| 1.6 Direct reuse | 22 |
| 1.7 Recycling | 25 |
| 2 Obsolescence | 0 |
| 2.1 Physical | 21 |
| 2.2 Functional | 20 |
| 2.3 Technical | 15 |
| 2.4 Economic | 6 |
| 2.5 Regulatory | 1 |
| 2.6 Aesthetic | 3 |
| 2.7 User-driven | 1 |
| 3 Enabling condition | 0 |
| 3.1 Component accessibility | 12 |
| 3.2 Disassembly | 24 |
| 3.3 Spare parts | 2 |
| 3.4 Repair information | 28 |
| 3.5 Skilled labour | 5 |
| 3.6 Service infrastructure | 16 |
| 3.7 Business model | 7 |
| 3.8 Regulation | 0 |
| 3.9 Certification | 3 |
| 3.10 User acceptance | 4 |
| 4 Negative mechanism | 0 |
| 4.1 Premature replacement | 22 |
| 4.2 Frequent upgrading | 2 |
| 4.3 Increased material complexity | 11 |

| | |
|---|---|
| 4.4 Additional transport | 2 |
| 4.5 Higher initial material use | 18 |
| 4.6 Increased comfort or use | 6 |
| 4.7 Burden shifting | 48 |
| 4.8 Lock-in | 15 |
| 4.9 Missing repair infrastructure | 15 |
| 5 Mitigation | 0 |
| 5.1 Design intervention | 30 |
| 5.2 Maintenance planning | 14 |
| 5.3 Supplier commitment | 1 |
| 5.4 User guidance | 0 |
| 5.5 Business-model intervention | 11 |
| 5.6 Policy intervention | 5 |
| 5.7 Lifecycle verification | 72 |

# 1 Circular strategy

Major code group for the type of circular intervention discussed in the source.

## 1.1 Circular strategy >> Maintenance

Actions that preserve the condition and performance of an existing window system without major replacement. Use when the source discusses regular inspection, cleaning, repainting, sealing, adjustment or upkeep. Exclude when the action replaces a major component or changes the system performance level. Example: periodic maintenance of wooden windows to prevent deterioration.

## 1.2 Circular strategy >> Repair

Corrective action that fixes damage or failure in an existing component. Use when a damaged part is fixed so the original window or component can remain in use. Exclude routine maintenance or full replacement. Example: repairing damaged seals, hinges, glazing beads or frame areas.

## 1.3 Circular strategy >> Component replacement

Replacement of a short-life or damaged part while retaining the larger window system. Use when the source discusses replacing seals, gaskets, glazing units, coatings, hardware or other parts. Exclude whole-window replacement. Example: replacing an IGU edge seal instead of discarding the entire glazing unit.

## 1.4 Circular strategy >> Upgrade

Improvement of performance while keeping most of the existing window system. Use when the source discusses energy retrofit, secondary glazing, storm windows or improved airtightness. Exclude full removal of the existing system. Example: adding secondary glazing to improve thermal performance.

## 1.5 Circular strategy >> Refurbishment

More extensive renewal of an existing window or component to restore function, appearance or performance. Use when the source discusses restoration, remanufacturing or returning a used component to serviceable condition. Exclude minor repair only. Example: restoring historic steel windows instead of replacing them.

## 1.6 Circular strategy >> Direct reuse

Use of an existing component again with little or no processing. Use when the source discusses reusing a component in the same or similar function. Exclude recycling, downcycling or material recovery. Example: reusing a recovered window frame or façade component.

## 1.7 Circular strategy >> Recycling

Processing material into secondary raw material after the component is no longer used. Use when the source discusses material recovery, recycling rates, recycling credits or end-of-life processing. Exclude intact component reuse. Example: recycling aluminium window frame material.

# 2 Obsolescence

Major code group for the reason a window system or component loses usefulness, value or acceptability.

## 2.1 Obsolescence >> Physical

Loss of function due to material ageing, damage, decay or wear. Use when deterioration, corrosion, seal failure, coating degradation or mechanical damage is the issue. Exclude market preference, regulation or outdated performance expectations. Example: timber decay, seal failure or corroded steel frames.

## 2.2 Obsolescence >> Functional

Loss of adequacy because the window no longer meets current performance needs. Use when the issue is thermal comfort, airtightness, acoustic performance, operability or energy performance. Exclude physical damage where performance remains adequate. Example: existing windows replaced because they are considered thermally inadequate.

## 2.3 Obsolescence >> Technical

Loss of usefulness because the design or technology prevents repair, upgrade or recovery. Use when components are bonded, inaccessible, inseparable or technically hard to disassemble. Exclude simple physical damage. Example: insulated glazing units cannot be refurbished because the edge seal is not replaceable.

## 2.4 Obsolescence >> Economic

Loss of usefulness because repair, reuse or maintenance is not financially attractive. Use when labour cost, service cost, storage cost, lack of market value or investment cost is the barrier. Exclude technical impossibility. Example: replacement chosen because repair appears too expensive or uncertain.

## 2.5 Obsolescence >> Regulatory

Loss of usefulness because the product no longer meets legal, safety, energy or certification requirements. Use when the source discusses regulation, certification, standards, warranty or compliance barriers. Exclude regulation mentioned only as general context. Example: reused components cannot be accepted because certification is missing.

## 2.6 Obsolescence >> Aesthetic

Loss of desirability because appearance, style or architectural preference changes. Use when replacement is driven by visual preference, façade image, modernization or perceived outdated appearance. Exclude actual damage or performance failure. Example: historic windows replaced because new windows are visually preferred.

## 2.7 Obsolescence >> User-driven

Loss of usefulness because user behaviour, expectations or comfort preferences change. Use when occupants, owners or users change demand for comfort, convenience, operation or appearance. Exclude changes caused mainly by technical failure. Example: users replace windows to achieve higher comfort even when repair is possible.

# 3 Enabling condition

Major code group for conditions that must exist for a circular strategy to work in practice.

## 3.1 Enabling condition >> Component accessibility

Components can be reached for inspection, maintenance, repair or replacement. Use when accessibility is necessary for the strategy to work. Exclude general repair without access issues. Example: seals and hardware are accessible for replacement.

## 3.2 Enabling condition >> Disassembly

The system can be taken apart without destroying valuable components. Use when the source discusses reversible connections, disassembly sequence, selective deconstruction or design for disassembly. Exclude demolition without recovery. Example: façade components removed in a planned sequence.

## 3.3 Enabling condition >> Spare parts

Replacement parts are available for maintenance, repair or component replacement. Use when the source discusses gaskets, seals, fittings, coatings or other parts needed over time. Exclude general material availability only. Example: spare seals available for future window repair.

## 3.4 Enabling condition >> Repair information

Information exists about materials, connections, maintenance history or repair procedure. Use when drawings, manuals, material passports, BIM data or disassembly guides are needed. Exclude repair that is possible without information barriers. Example: material passport records the window composition and connection types.

## 3.5 Enabling condition >> Skilled labour

Qualified people are available to inspect, repair, refurbish or disassemble the system. Use when expertise, workmanship or specialist knowledge affects the strategy. Exclude labour mentioned only as cost. Example: preservation professionals restore historic steel windows.

## 3.6 Enabling condition >> Service infrastructure

Local systems exist for repair, refurbishment, reuse, reverse logistics, storage or recycling. Use when the source discusses practical infrastructure outside the product itself. Exclude product design only. Example: local repair and refurbishment services support window service life extension.

### 3.7 Enabling condition >> Business model

A commercial or organisational model supports the circular strategy. Use when the source discusses service models, supplier responsibility, take-back, leasing, refurbishment services or market demand. Exclude technical feasibility only. Example: supplier offers long-term repair and component replacement.

### 3.8 Enabling condition >> Regulation

Policy, rules or public requirements support the strategy. Use when the source discusses policy support, legal requirements or regulatory drivers. Exclude regulation mentioned only in passing. Example: regulation encourages reuse or repair instead of disposal.

### 3.9 Enabling condition >> Certification

Certification, testing or warranty systems make reused or repaired components acceptable. Use when the source discusses certification barriers, performance validation or acceptance of secondary components. Exclude general quality only. Example: reclaimed façade parts need certification before reuse.

### 3.10 Enabling condition >> User acceptance

Users, owners or clients accept the circular strategy. Use when preferences, trust, appearance, comfort or perceived quality affect adoption. Exclude technical performance only. Example: owners accept repaired windows instead of demanding full replacement.

## 4 Negative mechanism

Major code group for mechanisms through which a circular strategy may create unintended consequences or reduce expected environmental benefit.

### 4.1 Negative mechanism >> Premature replacement

A window or component is replaced before the end of its useful service life. Use when repair, maintenance or lower-material intervention may have been sufficient. Exclude clearly necessary replacement due to severe failure. Example: full window replacement despite repairable defects.

### 4.2 Negative mechanism >> Frequent upgrading

Products are upgraded more often because better performance or new technology becomes available. Use when repeated upgrades may reduce the environmental benefit. Exclude one justified retrofit. Example: replacing windows repeatedly for marginal efficiency improvements.

## 4.3 Negative mechanism >> Increased material complexity

A strategy adds layers, materials or assemblies that make future repair, recycling or recovery harder. Use when coatings, laminated layers, adhesives, cladding or composite assemblies create complexity. Exclude simple material substitution without recovery implications. Example: added coatings improve performance but reduce recyclability.

## 4.4 Negative mechanism >> Additional transport

Circular strategy creates extra transport for repair, reuse, refurbishment, storage or recycling. Use when logistics may reduce expected benefit. Exclude negligible or irrelevant transport. Example: recovered components require long-distance transport to a reuse facility.

## 4.5 Negative mechanism >> Higher initial material use

A strategy requires more material at the beginning to improve durability, performance or future circularity. Use when added material may offset later savings. Exclude unchanged material use. Example: aluminium cladding reduces maintenance but increases production impact.

## 4.6 Negative mechanism >> Increased comfort or use

Efficiency or circular improvements lead to more use, higher comfort expectations or increased demand. Use when behaviour may offset environmental savings. Exclude comfort improvement without evidence of increased use. Example: better windows allow higher indoor temperatures and reduce heating savings.

## 4.7 Negative mechanism >> Burden shifting

Impact is moved to another lifecycle stage, actor, location or environmental category. Use when a benefit in one area causes harm elsewhere. Exclude simple uncertainty. Example: lower carbon impact comes with higher toxicity impact from coating choice.

## 4.8 Negative mechanism >> Lock-in

A long-lived solution restricts future repair, upgrading, disassembly or environmental improvement. Use when current design choices make later circularity harder. Exclude temporary barriers that can easily be changed. Example: permanent seals prevent future IGU refurbishment.

### 4.9 Negative mechanism >> Missing repair infrastructure

Repair, refurbishment or reuse fails because the needed services, skills or logistics are unavailable. Use when the strategy depends on a practical system that does not exist. Exclude barriers caused purely by product design. Example: repair is possible in theory but no local service provider exists.

## 5 Mitigation

Major code group for interventions that reduce rebound risk, burden shifting, implementation failure, lifecycle trade-offs or technological lock-in.

### 5.1 Mitigation >> Design intervention

A design change reduces the risk of unintended consequences. Use when the source suggests redesign, separable parts, reversible connections or accessible components. Exclude policy or business measures. Example: design the IGU edge seal as replaceable.

### 5.2 Mitigation >> Maintenance planning

Planned inspection and maintenance reduces premature failure or replacement. Use when schedules, inspections, defect classification or maintenance strategies are proposed. Exclude one-time repair only. Example: regular defect inspections for wooden windows.

### 5.3 Mitigation >> Supplier commitment

Suppliers or manufacturers support long-term repair, parts, take-back or product stewardship. Use when long-term responsibility is needed from the product provider. Exclude general manufacturer involvement. Example: manufacturer provides spare gaskets and repair guidance.

### 5.4 Mitigation >> User guidance

Users or owners receive instructions that support correct use, maintenance or repair decisions. Use when behaviour, acceptance or correct use matters. Exclude technical documentation aimed only at professionals. Example: owners receive guidance on when to repair instead of replace.

### 5.5 Mitigation >> Business-model intervention

A business model supports circular action and reduces practical barriers. Use when service contracts, leasing, repair services or take-back systems are proposed. Exclude product redesign alone. Example: maintenance service model supports window life extension.

## 5.6 Mitigation >> Policy intervention

Regulation, incentives or public requirements reduce barriers or improve adoption. Use when policy support is needed to make the circular strategy work. Exclude voluntary business action only. Example: policy supports repair, reuse certification or landfill diversion.

## 5.7 Mitigation >> Lifecycle verification

Environmental claims are checked through LCA, LCC, sensitivity analysis or scenario testing. Use when the source recommends verifying whether the expected benefit is real. Exclude circular benefit assumed without evidence. Example: compare maintenance and replacement scenarios before deciding.